



Aspen Exchanger Design and Rating Shell & Tube V14

TEMA Sheet



Heat Exchanger Specification Sheet

1	Company:									
2	Location:									
3	Service of Unit:					Our Reference:				
4	Item No.:					Your Reference:				
5	Date:		Rev No.:		Job No.:					
6	Size: 700 - 4500		mm		Type: BES Horizontal		Connected in: 1 parallel		1 series	
7	Surf/unit(eff.)		141.7		m ²		Shells/unit 1		Surf/shell(eff.) 141.7 m ²	
8	PERFORMANCE OF ONE UNIT									
9	Fluid allocation				Shell Side			Tube Side		
10	Fluid name				Methane fraction			Recycled water		
11	Fluid quantity, Total				kg/h			14849		
12	Vapor (In/Out)				kg/h			0 0		
13	Liquid				kg/h			312662 312662		
14	Noncondensable				kg/h			0 0		
15										
16	Temperature (In/Out)				°C			249.8 40.01		
17	Bubble / Dew point				°C			-218.53 / 96.08 -218.65 / 95.45		
18	Density Vapor/Liquid				kg/m ³			2.21 / 3.56 / 992.23		
19	Viscosity				cp			0.0176 / 0.0118 / 0.6514		
20	Molecular wt, Vap							15.97 15.64		
21	Molecular wt, NC							15.61 15.61		
22	Specific heat				kJ/(kg-K)			2.846 / 2.347 / 4.19		
23	Thermal conductivity				W/(m-K)			0.0688 / 0.0385 / 0.6315		
24	Latent heat				kJ/kg			2256 2407		
25	Pressure (abs)				kPa			600 586.124		
26	Velocity (Mean/Max)				m/s			9.31 / 16.07		
27	Pressure drop, allow./calc.				kPa			15 13.876		
28	Fouling resistance (min)				m ² -K/W			9E-05 0.00035 0.00044 Ao based		
29	Heat exchanged		3745		kW		MTD (corrected)		54 °C	
30	Transfer rate, Service		489.2		Dirty		650.4		Clean 995.7 W/(m ² -K)	
31	CONSTRUCTION OF ONE SHELL								Sketch	
32					Shell Side		Tube Side			
33	Design/Vacuum/test pressure:g		kPa		1000 / /		1000 / /			
34	Design temperature / MDMT		°C		300 / -37		50 / -37			
35	Number passes per shell				1		2			
36	Corrosion allowance		mm		3		3			
37	Connections		In mm		1 304.8 / 63 DIN		1 254 / 16 DIN			
38	Size/Rating		Out		1 304.8 / 63 DIN		1 254 / 16 DIN			
39	Nominal		Intermediate		/ 63 DIN		/ 16 DIN			
40	Tube #: 560		OD: 19		Tks. Average 2		mm		Length: 4500 mm Pitch: 23.75 mm Tube pattern:30	
41	Tube type: Plain		Insert:None		Fin#:		#/m		Material:SS 304	
42	Shell Carbon Steel		ID 700		OD 720		mm		Shell cover Carbon Steel	
43	Channel or bonnet		Carbon Steel						Channel cover Carbon Steel	
44	Tubesheet-stationary		Carbon Steel		-				Tubesheet-floating Carbon Steel	
45	Floating head cover		Carbon Steel						Impingement protection Square plate	
46	Baffle-cross Carbon Steel		Type Double segmental		Cut(%d)		23.56		VertiSpacing: c/c 250 mm	
47	Baffle-long -		Seal Type						Inlet 495.24 mm	
48	Supports-tube U-bend		0		Type					
49	Bypass seal		Tube-tubesheet joint		Expanded only (2 grooves)(App.A "i")					
50	Expansion joint -		Type None							
51	RhoV2-Inlet nozzle 1448		Bundle entrance 1366		Bundle exit 1099				kg/(m-s ²)	
52	Gaskets - Shell side		Flat Metal Jacket Fibe		Tube side		Flat Metal Jacket Fibe			
53	Floating head		Flat Metal Jacket Fibe							
54	Code requirements		ASME Code Sec VIII Div 1		TEMA class R - refinery service					
55	Weight/Shell		4483.3 Filled with water 6490.8		Bundle 2625.1				kg	
56	Remarks									
57										
58										



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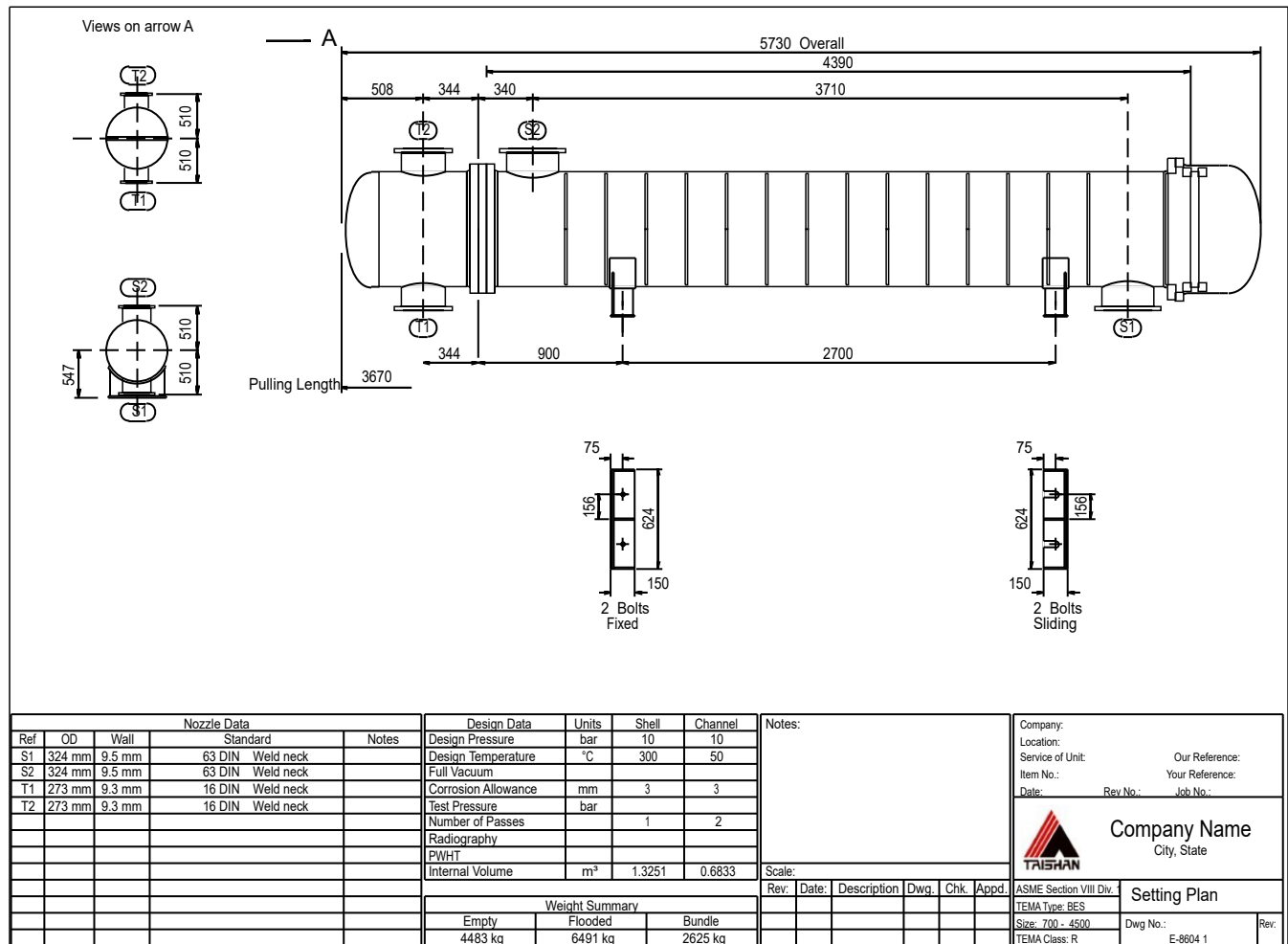
Overall Summary

1	Size	700	X	4500	mm	Type	BES	Hor	Connected in	1 parallel	1 series		
2	Surf/Unit (gross/eff/finned)			150.4	/	141.7	/		m² Shells/unit	1			
3	Surf/Shell (gross/eff/finned)			150.4	/	141.7	/		m²				
4	Rating / Checking	PERFORMANCE OF ONE UNIT											
5													
6	Process Data	Shell Side			Tube Side		Heat Transfer Parameters						
7	Total flow	kg/h		14849			312662	Total heat load			kW	3745	
8	Vapor	kg/h	14849	12530		0	0	Eff. MTD/ 1 pass MTD			°C	54 / 59	
9	Liquid	kg/h	0	2319		312662	312662	Actual/Reqd area ratio - fouled/clean			1.33	/ 2.04	
10	Noncondensable	kg/h		12347			0	Coef./Resist.			W/(m²-K)	m²-K/W %	
11	Cond./Evap.	kg/h		2319			0	Overall fouled			650.4	0.00154	
12	Temperature	°C	249.8	40.01		25	35	Overall clean			995.7	0.001	
13	Bubble Point	°C	-218.53	-218.65				Tube side film			6336.4	0.00016 10.26	
14	Dew Point	°C	96.08	95.45				Tube side fouling			2255.6	0.00044 28.83	
15	Vapor mass fraction		1	0.8438		0	0	Tube wall			6669	0.00015 9.75	
16	Pressure (abs)	kPa	600	586.124		690	662.474	Outside fouling			11111.1	9E-05 5.85	
17	DeltaP allow/cal	kPa	15	13.876		50	27.526	Outside film			1435.7	0.0007 45.3	
18	Velocity	m/s	14.41	7.52		1.73	1.8						
19	Liquid Properties							Shell Side Pressure Drop			kPa	%	
20	Density	kg/m³		992.23		997.15	994.11	Inlet nozzle			1.534	11	
21	Viscosity	cp		0.6514		0.8904	0.7185	InletspaceXflow			0.792	5.67	
22	Specific heat	kJ/(kg-K)		4.19		4.313	4.314	Baffle Xflow			6.535	46.83	
23	Therm. cond.	W/(m-K)		0.6315		0.6114	0.6251	Baffle window			4.076	29.21	
24	Surface tension	dynes/cm		69.5				OutletspaceXflow			0.457	3.28	
25	Molecular weight			18.02		18.02	18.02	Outlet nozzle			0.561	4.02	
26	Vapor Properties							Intermediate nozzles					
27	Density	kg/m³	2.21	3.56				Tube Side Pressure Drop			kPa	%	
28	Viscosity	cp	0.0176	0.0118				Inlet nozzle			1.362	4.95	
29	Specific heat	kJ/(kg-K)	2.846	2.347				Entering tubes			1.554	5.65	
30	Therm. cond.	W/(m-K)	0.0688	0.0385				Inside tubes			21.453	77.96	
31	Molecular weight		15.97	15.64				Exiting tubes			2.372	8.62	
32	Two-Phase Properties							Outlet nozzle			0.775	2.82	
33	Latent heat	kJ/kg	2256	2407	Intermediate nozzles								
34	Heat Transfer Parameters							Velocity / Rho*V2			m/s	kg/(m-s²)	
35	Reynolds No. vapor		34283.75	43025.23				Shell nozzle inlet			25.62	1448	
36	Reynolds No. liquid			144.47		29049.8	37311.1	Shell bundle Xflow			14.41 7.52		
37	Prandtl No. vapor		0.73	0.72				Shell baffle window			16.07 8.38		
38	Prandtl No. liquid			4.32		6.28	4.96	Shell nozzle outlet			13.42	759	
39	Heat Load			kW				Shell nozzle interm					
40	Vapor only			-1664.9			0				m/s	kg/(m-s²)	
41	2-Phase vapor			-492.5			0	Tube nozzle inlet			1.71	2923	
42	Latent heat			-1485.8			0	Tubes			1.73 1.8		
43	2-Phase liquid			-101.8			0	Tube nozzle outlet			1.72	2932	
44	Liquid only			0				Tube nozzle interm					
45	Tubes					Baffles		Nozzles: (No./OD)					
46	Type			Plain	Type	Double segmental				Shell Side		Tube Side	
47	ID/OD	mm	15 / 19		Number	14	Inlet	mm	1 / 323.85	1 / 273.05			
48	Length act/eff	m	4.5 / 4.2405		Cut(%d)	23.56	14.69	Outlet	1 / 323.85	1 / 273.05			
49	Tube passes		2		Cut orientation	V	Intermediate	/	/				
50	Tube No.		560		Spacing: c/c	mm	250	Impingement protection	Square plate				
51	Tube pattern		30		Spacing at inlet	mm	495.24						
52	Tube pitch	mm	23.75		Spacing at outlet	mm	495.24						
53	Insert	None											
54	Vibration problem (HTFS / TEMA)	Possible	/	TEMA RhoV2 limit exceeded							No		



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Setting Plan





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Tubesheet Layout

